

Outline

Clinical Standards
of Wearable
Devices (PA)

Devices (Actigraphy)

Data

Typical Analysis

Analysis

Central Tendency

Measures of Variance

Standardized Measures

Comparison Tests

Analysis of Variance

Wearable Technologies and Applications (Wearable Informatics)

Winfree

Lecture 4

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Clinical and Research devices often require wired connections, but wireless is taking over as security is

addressed.

ActiGraph.

Hardware Product Evolution



Sometimes the data are stored in a mixed format, text and low level binary

Log Record Types

Note that some undocumented records are used for internal state or testing. They may be safely ignored.

ID (dec)	ID (hex)	Type	Description
0	0x00	ACTIVITY	One second of raw activity samples packed into 12-bit values in XYZ order.
2	0x02	BATTERY	Battery voltage in millivolts as a little-endian unsigned short (2 bytes).
3	0x03	EVENT	Logging records used for internal debugging. These records notably contain information about idle sleep mode. When entering idle sleep mode, a record with payload 0x08 is created. When exiting idle sleep mode, a record with payload 0x09 is created.
4	0x04	HEART_RATE_BPM	Heart rate average beats per minute (BPM) as one byte unsigned integer.
5	0x05	LUX	Lux value as a little-endian unsigned short (2 bytes).
6	0x06	METADATA	Arbitrary metadata content. The first record in every log is contains subject data in JSON format.
7	0x07	TAG	13 Byte Serial, 1 Byte Tx Power, 1 Byte (signed) RSSI
9	0x09	EPOCH	60-second epoch data
11	0x0B	HEART_RATE_ANT	Heart Rate RR information from ANT+ sensor.
12	0x0C	EPOCH2	60-second epoch data
13	0x0D	CAPSENSE	Capacitive sense data
14	0x0E	HEART_RATE_BLE	Bluetooth heart rate information (BPM and RR). This is a Bluetooth standard format.
15	0x0F	EPOCH3	60-second epoch data

Typically time based

- ▶ A header often defines parameters
- ▶ Each row is an entry
- ▶ Each column is a measure

Typical Analysis

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Analysis of Variance

- ▶ Measures of central tendency
- ▶ Measures of variability
- ▶ Comparisons
- ▶ Trends (Binned, Change Point, Detrended Flucuation)

Central Tendency

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- ▶ Arithmetic Mean:

$$\bar{A} = \frac{1}{n} \sum_{i=1}^n a_i$$

- ▶ Geometric Mean¹:

$$\bar{G} = (\prod_{i=1}^n x_i)^{\frac{1}{n}} = \sqrt[n]{x_1 x_2 \dots x_n}$$

- ▶ Harmonic Mean²:

$$\bar{H} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \dots + \frac{1}{x_n}} = \frac{n}{\sum_{i=1}^n \frac{1}{x_i}}$$

- ▶ Median

- ▶ Mode

¹useful whenever the quantities to be averaged combine multiplicatively

²most appropriate average for ratios and rates such as speeds, robust to outliers

Measures of Variance

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Standardized Measures (Z-Score)

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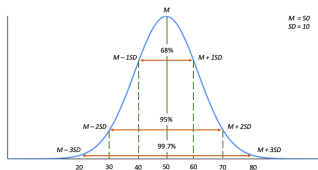
Comparison Tests

Analysis of Variance

What if we wanted to make some comparisons when measures are in different units? Or what if the samples seem to have some bias, and we want to understand deviations from that bias?

$$Z_i = \frac{x_i - \mu}{\sigma}$$

Standard deviations in a normal distribution



Comparison Tests

pre post
two groups

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Anlaysis of Variance (ANOVA)

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